

Reproducing the D-CBG baselines

arXiv:2412.10193v3 (Schiff et al., ICLR 2025) on QM9

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1 What is being reproduced, and what is ours

D-CBG needs a *noisy classifier* per property, and the paper releases none. All four classifiers used here (`{qed,ring_count}` $\times$ `{uniform,absorbing_state}`) are therefore our own training runs. On the generative side the split is:

- **UDLM** — the released `kuleshov-group/udlm-qm9` checkpoint, i.e. the paper’s own model. This is the arm that has to match.
- **MDLM** — our own 25k-step run; HuggingFace publishes no MDLM-QM9.

So even the UDLM arm is not a pure re-run of the paper’s pipeline: the classifier differs. This matters when reading the z column, because the paper’s $\pm$ is over *five sampling seeds on one checkpoint* and contains no training variation of either component.

Two conversions are needed before the numbers are comparable. The paper reports `NUM. VALID / NUM. NOVEL` as counts out of 1,024 and its `MEAN` column over *novel* sequences only, while `our_qm9_eval.py` writes `VALID` as a fraction of 1,024 and `NOVEL` as a fraction of the valid set. And we contribute one seed against a five-seed mean, so the spread of the difference is $\sqrt{1 + 1/5} \sigma$, not σ ; the z below carries that factor. The property σ is floored at 0.005 because the paper prints that column to two decimals, so a printed “0.00” only means < 0.005 .

2 The four rows that appear in Table 5

The main paper’s Table 5 does not state which γ or `use_approx` produced each row; the appendix does (Tables 20, 23, 29, 32). Those are the settings reproduced here.

3 Where the residual gap comes from

Reading the two tables together:

1. **The MDLM approximate arm on QED** is the one real offset: mean signed z of $+4.3/+3.9$ on valid/novel with the property at -0.1 . Our checkpoint is *more robust* to the first-order

property	model	setting	valid	z	novel	z	property	z
QED	MDLM	$\gamma = 3$, exact	390 / 417.6	-1.3	100 / 116.6	-1.7	0.588 / 0.58	+1.4
QED	UDLM	$\gamma = 10$, exact	989 / 994.8	-1.9	63 / 63.8	-0.1	0.609 / 0.61	-0.2
ring count	MDLM	$\gamma = 10$, approx.	88 / 113.0	-2.6	67 / 85.6	-1.9	4.597 / 4.75	-0.6
ring count	UDLM	$\gamma = 8$, exact	910 / 897.2	+1.0	449 / 432.0	+0.8	4.837 / 4.84	-0.1

Each metric column reads *ours* / *paper*; z is $(\text{ours} - \text{paper}) / \sqrt{1 + 1/5} \sigma_{\text{paper}}$.

Table 1: The four D-CBG rows of the paper’s Table 5. Three are within 2σ on all three metrics; MDLM ring count is outside on the two count metrics and matches on the property.

property	model	arm	n	mean z valid	mean z novel	mean z property
QED	MDLM	exact	5	-0.0	-0.4	-0.6
QED	MDLM	approx.	5	+4.3	+3.9	-0.1
QED	UDLM	exact	5	-0.0	-0.3	+0.1
QED	UDLM	approx.	5	+1.5	-1.0	+0.2
ring count	MDLM	exact	2	+4.4	+3.1	-1.4
ring count	MDLM	approx.	6	+1.1	-0.2	+0.1
ring count	UDLM	exact	6	+0.9	-0.8	-0.6
ring count	UDLM	approx.	6	-0.1	-1.1	+0.6

The property column is centred everywhere ($|\cdot| \leq 0.7$) except the $n=2$ collapsed MDLM ring exact arm. The deviations live in validity and novelty counts, not in how hard the guidance steers.

Table 2: Mean *signed* z per arm. Random scatter would average to zero; a consistent sign is a systematic offset.

approximation than theirs — same property at higher validity. Expected, since MDLM-QM9 is our own training run.

2. **Everything else is centred.** In particular UDLM exact, the arm that must match, is within $|z| \leq 1$ on average for both properties.
3. Individual cells still scatter, worst at UDLM ring $\gamma=3$ exact (4.56 against 4.70 ± 0.03). A 0.6% printed σ is tight enough that any pipeline difference shows up as a large z .

Verdict. The baselines reproduce. Absolute numbers can be quoted against the paper; frontier comparisons within a fixed checkpoint are safer still, and are what the sweep reports do.

γ	arm	valid	paper	z	novel	paper	z	QED	paper	z
1	exact	537	526.4	+0.6	189	170.8	+1.1	0.557	0.56	-0.6
2	exact	488	524.8	-1.5	127	139.0	-0.9	0.574	0.58	-1.0
3	exact	390	417.6	-1.3	100	116.6	-1.7	0.588	0.58	+1.4
4	exact	197	200.4	-0.3	56	66.4	-1.2	0.566	0.58	-2.5
5	exact	33	24.6	+2.4	14	11.6	+1.0	0.575	0.58	-0.4
1	approx.	504	476.0	+1.7	208	221.4	-1.6	0.467	0.46	+0.6
2	approx.	400	363.4	+1.9	165	172.4	-0.7	0.464	0.47	-0.5
3	approx.	307	244.0	+4.3	145	114.0	+4.6	0.472	0.47	+0.2
5	approx.	187	120.6	+9.1	80	48.4	+5.3	0.488	0.50	-1.1
10	approx.	108	72.6	+4.5	45	21.2	+12.1	0.556	0.55	+0.3

Table 3: MDLM, QED: every γ we ran against the paper. 5/10 rows are within 2σ on all three metrics.

γ	arm	valid	paper	z	novel	paper	z	QED	paper	z
1	exact	949	933.4	+1.9	118	134.6	-2.1	0.528	0.53	-0.2
2	exact	903	911.2	-0.9	115	119.8	-0.3	0.567	0.57	-0.3
3	exact	946	941.0	+0.4	102	96.4	+0.6	0.588	0.58	+0.7
5	exact	968	967.6	+0.2	82	77.8	+0.4	0.593	0.59	+0.3
10	exact	989	994.8	-1.9	63	63.8	-0.1	0.609	0.61	-0.2
1	approx.	997	996.4	+0.1	121	132.2	-1.2	0.477	0.47	+0.7
2	approx.	976	974.8	+0.1	116	110.6	+0.5	0.485	0.49	-0.5
3	approx.	955	925.0	+3.4	95	111.4	-1.6	0.523	0.51	+1.1
5	approx.	856	828.2	+1.4	78	86.2	-0.6	0.529	0.53	-0.1
10	approx.	819	783.2	+2.4	59	67.0	-2.4	0.556	0.56	-0.3

Table 4: UDLM, QED: every γ we ran against the paper. 7/10 rows are within 2σ on all three metrics.

γ	arm	valid	paper	z	novel	paper	z	ring count	paper	z
1	exact	188	143.8	+6.1	149	121.2	+3.6	4.745	5.00	-3.3
2	exact	2	0.4	+2.7	2	0.4	+2.7	6.000	3.00	+0.6
1	approx.	493	455.4	+1.4	209	229.4	-1.2	2.809	2.52	+2.4
2	approx.	352	327.6	+1.4	176	176.4	-0.0	2.966	2.82	+0.9
3	approx.	267	223.8	+2.6	150	136.0	+1.3	3.400	3.25	+0.5
5	approx.	178	135.0	+4.0	111	94.4	+1.4	3.856	4.11	-0.9
8	approx.	117	121.2	-0.3	86	92.6	-0.5	4.244	4.54	-1.7
10	approx.	88	113.0	-2.6	67	85.6	-1.9	4.597	4.75	-0.6

Table 5: MDLM, ring count: every γ we ran against the paper. 2/8 rows are within 2σ on all three metrics.

γ	arm	valid	paper	z	novel	paper	z	ring count	paper	z
1	exact	807	797.4	+0.9	275	279.0	-0.2	4.033	4.12	-2.0
2	exact	830	829.4	+0.0	307	336.4	-2.5	4.619	4.54	+2.4
3	exact	896	862.6	+3.9	345	363.8	-1.3	4.557	4.70	-4.4
5	exact	885	889.2	-0.3	400	404.0	-0.4	4.695	4.76	-2.0
8	exact	910	897.2	+1.0	449	432.0	+0.8	4.837	4.84	-0.1
10	exact	892	891.6	+0.0	416	431.4	-1.2	4.885	4.76	+2.3
1	approx.	992	978.4	+3.2	154	166.0	-0.9	2.519	2.49	+0.4
2	approx.	901	892.2	+0.7	147	171.2	-1.6	3.184	3.09	+1.1
3	approx.	736	763.8	-2.2	197	198.8	-0.2	3.893	3.76	+1.4
5	approx.	712	726.8	-0.9	223	245.8	-3.0	4.435	4.47	-0.5
8	approx.	794	796.6	-0.1	298	304.4	-0.6	4.661	4.66	+0.0
10	approx.	792	816.6	-1.5	355	359.8	-0.2	4.780	4.70	+1.5

Table 6: UDLM, ring count: every γ we ran against the paper. 6/12 rows are within 2σ on all three metrics.